Diego **Zamboni**

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CISO • Organizational Leader • Security Expert • Computer Scientist

Senior computer scientist, security expert and organizational leader with 30 years of experience across security governance, cloud security architecture, research and engineering. I combine strategic leadership, technical depth and clear communication to help organizations design secure systems, scale teams and deliver measurable outcomes.

Professional Highlights

Management and leadership, IT security, cloud computing

- Chief Information Security Officer for Governance at Avaloq, defining and managing Avaloq's global ISO27001-certified Information Security Management System
- Managed security architecture at the Stellantis *Virtual Engineering Workbench* project. Worked with Stellantis CISO and business stakeholders to define governance, establish security best practices and drive risk analysis, threat modeling and mitigation.
- Established scalable and durable mechanisms to enable partners to work securely in the Volkswagen Digital Production Platform (DPP) program.
- Managed security architecture, risk management, data governance and compliance (ISO27001, ISAE3402/3000, etc.) for Swisscom's Cloud platforms.
- Established and led the Swisscom IT Clouds security community of practice.
- Established and led the *Health and State Management* team at Swisscom to design, implement and operate a framework for scalable monitoring, logging and alerting for Swisscom's Cloud platforms.
- Established and led the first computer security organization at UNAM, which has grown into the university's Information Security Coordination (UNAM-CERT).
- Managed IT security customer relationships at HP Enterprise Services, including overseeing the activities of operational and engineering teams, risk and compliance management, requirements discussion and reporting.

Research, architecture and design

- Designed the *Orchard* monitoring framework for Swisscom's *Application Cloud* platform, and led the team that implemented it and brought it into production.
- Designed and implemented the *Billy Goat* malware capture and analysis system at IBM.

Experience

Avaloq — *Switzerland*
CISO Governance — *Jan 2024 - Present*

Lead Avaloq's global CISO Governance team and define and monitor the company's Information Security Management System.
- Own security policy and governance direction.
- Provide second-line oversight of first-line security functions.
- Drive compliance monitoring and control effectiveness in a regulated fintech environment.

Amazon Web Services — *Switzerland*
Global Security Architect / Senior Global Security Architect — *Sep 2021 - Dec 2023*

Worked with global AWS customers and internal teams to improve security posture, define secure architecture patterns and scale security awareness.
- Led security architecture in the Stellantis Virtual Engineering Workbench program.
- Established security workstreams, metrics and reporting for business and technical stakeholders.
- Helped launch and scale Security Champions and Security Guardians initiatives.
- Supported secure collaboration mechanisms for the Volkswagen Digital Production Platform.

Swisscom — *Switzerland*
Enterprise Architect and IT Clouds Solution Security Architect — *Apr 2019 - Sep 2021*

Led security architecture, risk and compliance activities for Swisscom cloud platforms across multiple services.
- Directed compliance and governance activities including ISO27001, ISAE3402/3000 and GDPR requirements.
- Built and led a Security Champions community across IT Clouds teams.
- Coordinated threat modeling, audits, penetration testing and vulnerability management.

Team Lead & Product Owner for Health & State Management — *Aug 2014 - Apr 2019*

Built and led the Health and State Management team for Swisscom cloud platforms.
- Owned roadmap and prioritization with product managers and stakeholders.

- Defined and delivered platform-wide monitoring, logging and alerting capabilities.
- Led the *Orchard* project through its implementation, production release and further improvements and development.

CFEngine AS — *Norway/U.S.A. (remote)*
PRODUCT MANAGER — *Oct 2011 - Jun 2014*

- CFEngine Advocate, with a special focus on security.
- Managed the CFEngine language roadmap.
- Created and led the CFEngine Design Center project, which was the foundation for the current CFEngine Build service.
- Coordinated the work on CFEngine third-party integration (e.g. AWS EC2, VMware, Docker and OpenStack).
- Wrote the book *Learning CFEngine 3*, published by O'Reilly Media, which became the de facto introductory text to CFEngine.

IBM Zurich Research Lab — *Switzerland*
RESEARCH STAFF MEMBER — *Oct 2001 - Oct 2009*

Member of IBM Zurich's Global Security Analysis Laboratory, working on intrusion detection, malware containment and virtualization security research.

National Autonomous University of Mexico (UNAM) — *Mexico*
FOUNDER AND LEAD OF COMPUTER SECURITY AREA — *Aug 1995 - Aug 1996*

- Established UNAM's *Computer Security Area*, the University's first team dedicated to computer security, which has evolved into the *Information Security Organization (UNAM-CERT)*.
- Managed up to nine people working on different projects related to computer security.
- Provided security services to the whole University, including incident response, security information, auditing and teaching.

Education

Purdue University — *West Lafayette, IN, U.S.A.*
PH.D. IN COMPUTER SCIENCE — *Aug 1996 - Aug 2001*

- Thesis: Using Internal Sensors for Computer Intrusion Detection. Advisor: Eugene H. Spafford

Purdue University — *West Lafayette, IN, U.S.A.*
M.S. IN COMPUTER SCIENCE — *Aug 1996 - May 1998*

- Advisor: Eugene H. Spafford

National Autonomous University of Mexico (UNAM) — *Mexico City, Mexico*
BACHELOR'S DEGREE IN COMPUTER ENGINEERING — *Aug 1989 - Jul 1995*

- Thesis: UNAM/Cray Project for Security in the Unix Operating System (in Spanish, original title: *Proyecto UNAM/Cray de Seguridad en el Sistema Operativo Unix*)

Certifications

See full list on Credly (www.credly.com/users/zzamboni)

Certified Information Systems Security Professional (CISSP) — *Apr 2019*
ISC2

Selected Research

See the *online publications list* (zzamboni.org/vita/publications) for publication details.

Billy Goat: Active worm detection and capture — *2002 - 2008*
IBM RESEARCH

- Pioneered active worm-capture technology that became the foundation for modern honeypots and honeynets
- Designed system to simulate thousands of vulnerable hosts to attract and capture propagating worms
- Implemented automated analysis to extract signatures and update intrusion detection/prevention systems
- Publications: [18], [25]

Ph.D. Thesis: Using internal sensors and embedded detectors for intrusion detection — *1996 - 2001*
PURDUE UNIVERSITY

- Investigated novel approaches to data collection for intrusion detection systems
- Implemented and evaluated internal sensors and embedded detectors as data collection mechanisms
- Analyzed trade-offs and effectiveness of different sensor placement strategies in IDS architectures
- Publications: [11], [19], [20], [27], [26], [32]

Software

Open-source software projects

- GitHub: github.com/zzamboni
- GitLab: gitlab.com/zzamboni

Honors & Awards

May 2020 **IEEE Security & Privacy Test of Time Award (IEEE S&P page, CERIAS blog post)**, IEEE

2010 **CFEngine Champion**, CFEngine AS

May 1996 **Fulbright Scholarship (for pursuing Ph.D. studies at Purdue University)**, Fulbright Program and CONACYT

Selected publications

Full list online (zzamboni.org/vita/publications)

Books

[1] Diego Zamboni. *Publishing with Emacs, Org-mode and Leanpub*. Leanpub, June 2020.

[2] Diego Zamboni. *Learning CFEngine*. O'Reilly Media, Inc. 2012–2017, afterwards self-published, 2012.

Editorial Activities

[3] Diego Zamboni and Christopher Kruegel, editors. *Proceedings of the Recent Advances in Intrusion Detection (RAID): 9th International Symposium*. Lecture Notes in Computer Science. Secaucus, NJ, USA: Springer-Verlag New York, Inc., 2006.

Theses

[4] Diego Zamboni. "Using Internal Sensors for Computer Intrusion Detection". CERIAS TR 2001-42. PhD thesis, West Lafayette, IN: Purdue University, August 2001.

Refereed Papers

[5] James Riordan, Diego Zamboni, and Yann Duponchel. "Building and deploying Billy Goat, a worm-detection system". In: *Proceedings of the 18th Annual FIRST Conference*. June 2006.

[6] Florian Kerschbaum, Eugene H. Spafford, and Diego Zamboni. "Using internal sensors and embedded detectors for intrusion detection". In: *Journal of Computer Security* 10.1,2 (2002), pages 23–70.

[7] Christoph L. Schuba, Ivan V. Krsul, Markus G. Kuhn, Eugene H. Spafford, Aurobindo Sundaram, and Diego Zamboni. "Analysis of a Denial of Service Attack on TCP". In: *Proceedings of the 1997 IEEE Symposium on Security and Privacy*. Awarded the 2020 IEEE Security & Privacy Test of Time Award. IEEE Computer Society, IEEE Computer Society Press, May 1997. Pages 208–223.

Skills

Leadership 32 years of multidisciplinary team and project leadership experience · IT Enterprise Architecture · Scaled Agile Framework (SAFe)

Information and Cyber Security Enterprise security governance · Enterprise security architecture · Virtualization and cloud computing security · Risk management and compliance · Intrusion detection and prevention · Software security and secure software development · ISO27001

Technology Broad and deep IT expertise · Cloud computing · Computer security · Operating systems · Networking · Configuration management · Software & services development · Programming languages

Research Ph.D. in Computer Science · 9 years of experience at IBM Research

Communication Excellent written and spoken communication skills · Extensive public speaking experience · Professional writing and teaching experience

Languages

Spanish Native
English Full proficiency
German Intermediate proficiency (B2 level)

References

Available by request